Active Maintenance Report: green_etal_2016

Green, Krasno, Coppock, Farrer, Lenoir and Zingher (2016, Electoral Studies): The Effects of Lawn Signs on Vote Outcomes

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This repository holds the actively maintained replication code for Green, Krasno, Coppock, Farrer, Lenoir and Zingher (2016), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1016/j.electstud.2015.12.002
Replication archive	10.7910/DVN/K2TLDB
Pre-analysis plan (Experiment 3)	osf.io/umysq
Pre-analysis plan (Experiment 4)	osf.io/ejfgk

The data are not redistributed here. The deposit is 12 MB across 33 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. It also holds `in_text_claims.R`, which recomputes every quantity the article states and prints it beside the sentence that states it. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of every number the article and appendix print, `archive_values.csv` is what the deposited scripts themselves print, `build_ground_truth.R` assembles the comparison, and `claims_gate.R` enforces the coverage. `errata.qmd` corrects the published article and appendix, rendering to `green_etal_2016_errata.pdf` at the root and writing its entries to `errata_entries.csv`. `original/` is created by the download script and is deliberately absent from the repository. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `green_etal_2016.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 33 files, and produces every table and figure into `maintained/output/`. Required packages: tidyverse, estimatr, metafor, gridExtra, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about two minutes, almost all of it in the randomization inference, which

refits 120,000 pairs of weighted models across 40,000 permutations. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Yes. All six analysis scripts execute without error on a current R installation, with no hardcoded paths, no functions called before they are defined, and no deprecated calls that have since been removed. Every package they name still installs from CRAN, including `beepr` 2.0, which only one of the random-assignment scripts calls, to play a sound on completion.

Running is not the same as reproducing, and one line changes a great many numbers. `Experiment_2_Analysis.R` picks the observed treatment assignment out of a matrix of 10,000 admissible randomizations with `set.seed(12345); sample(1:10000, 1)`. R 3.6.0 changed how `sample()` converts uniform draws to integers, so that line returned column 7210 when the paper was written and returns column 8243 today. Column 7210 is the assignment that was actually deployed, and it is the one the archive's own pre-saved `exp_2.RData` encodes, so the archive contains the right answer and the live script path no longer finds it. 43 of the 378 published values a deposited script can be checked against therefore fail to reproduce, and all 43 come from that one line: every cell of Table 4, every cell of appendix Table C.2, and the Experiment 2 randomization inference p-value. The deposit's other scripts read `exp_2.RData` directly and are unaffected, which is why Table 7, Table D.5 and the pooled quantities reproduce exactly.

1.2 Does the maintained rewrite reproduce the paper?

Yes, without exception. All 400 published values that can be compared against the rewrite match to the precision the page prints, including every panel label of Figure 2 and all 345 cells of the 12 published floats. A further 7 claims have no printed number to compare, either because the sentence states a shape rather than a value or because the deposit draws the quantity without a seed; 7 of 7 hold against the pipeline's own estimates. The remaining 74 published numbers are design facts the deposit does not record, such as how many signs were planted and how far in advance, and are verified against the article's own methods section rather than against code.

That agreement is between the page and the deposit, and it does not make the deposit right. The deposited Experiment 1 script sums vote shares across merged precincts and the printed

numbers carry the result, so 38 of the matches above are matches to an error, and each carries `defect_locus = archive`. Reading the article and its appendix word by word found further errors of description and wording. `green_etal_2016_errata.pdf`, at the root of this repository, lists 23 corrections and one unresolved citation, set out under Errata below. None changes the paper's conclusions.

The rewrite reaches Experiment 2 through the pre-saved `exp_2.RData` object rather than through the seed, which is what makes it immune to the R version change. Reaching it any other way is a live hazard rather than a hypothetical one: until August 2026 the rewrite's randomization inference script took Experiment 2's observed assignment from its own cleaning script, which redraws the seed, and consequently reported 0.79 where the article reports 0.90. That script now reads the same canonical object as everything else.

2 Paper overview

Citation: Green, D. P., Krasno, J. S., Coppock, A., Farrer, B. D., Lenoir, B. and Zingher, J. N. (2016). "The effects of lawn signs on vote outcomes: Results from four randomized field experiments." *Electoral Studies*, 41, 143-150. DOI: 10.1016/j.electstud.2015.12.002

Summary: Four randomized field experiments test whether lawn signs change vote outcomes. In each, geographic units (electoral districts or precincts) were assigned to receive signs, to be adjacent to a unit that received signs, or to neither, so that each experiment estimates a direct effect and a spillover effect at once. Because adjacency makes the assignment probabilities unequal across units, every estimate is inverse-probability weighted, and inference is randomization based: an observed F-statistic for the joint null of no direct and no indirect effect is compared against 10,000 pre-computed admissible randomizations. The four experiments are a 2012 congressional general election in New York, a 2013 mayoral primary in Albany, a 2013 gubernatorial general election in Virginia, and a 2015 county commissioner primary in Cumberland County, Pennsylvania. Pooling the four by fixed-effects meta-analysis gives a direct effect of 1.7 percentage points of vote share (SE 0.7) and a spillover effect of 1.5 points (SE 0.6).

3 Original archive reproducibility

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

Script	Status on current R	Resolution
Lawn_Signs_Source.R	Clean (sourced by all scripts)	No changes required
Experiment_1_Analysis.R	Clean	No changes required
Experiment_2_Analysis.R	Runs clean, but Table 4 and the Experiment 2 p-value no longer reproduce	set.seed(12345, sample.kind = 'Rounding'), or read the deposited exp_2.RData
Experiment_3_Analysis.R	Clean	No changes required

Experiment_4_Analysis.R	Clean	No changes required
Pooled_analysis.R	Clean	No changes required
in_text_calculations.R	Clean	No changes required
Experiment_1_RA_Function.R	Parses; regenerates a deposited matrix, not re-run	None needed: its output is deposited
Experiment_2_RA_Function.R	Parses; regenerates a deposited matrix, not re-run	None needed: its output is deposited
Experiment_3_RA_Function.R	Parses; regenerates a deposited matrix, not re-run	None needed: its output is deposited
Experiment_4_RA_Function.R	Parses; needs beepr, which installs from CRAN	install.packages('beepr')

Every package the archive names still installs from CRAN: `stargazer`, `xtable`, `sandwich`, `lmtest`, `ggplot2`, `dplyr`, `randomizr`, `beepr` and `rmeta`. The last of these is the fragile one. `rmeta` 3.0 has not been updated since March 2018 and survives on CRAN by its archiving policy rather than by maintenance, and `Pooled_analysis.R` depends on it for a single call to `meta.summaries()`. The maintained rewrite pools with `metafor::rma(method = "FE")` instead, so it does not inherit the exposure.

The four `*_RA_Function.R` scripts regenerate the permutation matrices from scratch. They parse cleanly and their packages install, but they were not re-run to completion: they are stochastic, they take a long time, and the matrices they build are themselves deposited, so re-running them would replace a deposited object with a different one rather than check anything.

3.1 The Experiment 2 seed

The one substantive failure has nothing to do with packages. `Experiment_2_Analysis.R` opens by drawing the observed treatment assignment out of the deposited permutation matrix:

```
set.seed(12345)
chosen.rand <- sample(1:10000, 1)
```

R 3.6.0, released in April 2019, replaced the "Rounding" method that `sample()` used to convert uniform variates to integers with a "Rejection" method. The same seed therefore selects a different column than it did when the analysis was run: 7210 under the old method, 8243 under the new one. The two columns are different treatment assignments and give different answers.

Column 7210 is the assignment the experiment actually used. Three independent checks agree on that: it is the column the archive's own pre-saved `exp_2.RData` encodes, it is the column whose treated set matches the authors' record of which districts received signs, and it is what `set.seed(12345, sample.kind = "Rounding")` returns on any R version. Column 7210 also reproduces the published Table 4 and the published Experiment 2 p-value; column 8243 reproduces neither.

Table 2: What the Experiment 2 permutation column decides. The published column is the article's own string; the other two are computed.

Quantity	Published	Column 7210 (R < 3.6)	Column 8243 (R 3.6+)
Table 4 M1 treated	0.009	0.009	0.130
Table 4 M2 treated	-0.014	-0.014	0.019
Table C.2 margin M1 treated	24.673	24.673	21.495
Table C.2 turnout M2 treated	9.029	9.029	5.985
Section 5.2 p-value	0.90	0.90	0.79

The archive is not wrong. It contains both the script and the object the script was meant to produce, and the object is correct. What it lacks is any signal that the two have come apart, which is the failure this maintenance program exists to catch: a reader running the deposited script today gets numbers that differ from the paper, with no error and no warning.

4 The extraction and the two instruments

Everything below rests on one hand-reviewed file. `ground_truth/published_claims.csv` is the **extraction**: every numeric token in the article and the online appendix, read off the published pages, classified by hand, and committed. It carries 481 rows. Table cells, figure labels, prose quantities, design facts and the numbers in appendix A that come from election returns rather than from the deposit are all in it, each with the string the page prints and the number of decimals the page prints it at.

Table 3: The extraction by class. A pipeline claim is one the code should produce; a descriptive claim states a shape or a count rather than a value; definitional, structural and transcribed claims are design facts, page structure, and numbers taken from outside the deposit.

Class	Rows	Needing a claim block
pipeline	388	388
definitional	49	14
structural	22	0
transcribed	17	0
descriptive	5	5

Two instruments then read the pipeline's output independently. `ground_truth/build_ground_truth.R` reshapes each output file into a long table keyed by claim identifier and joins it to the extraction. `maintained/in_text_claims.R` selects the same numbers cell by cell, quoting the article's own sentence beside each prose claim, and prints one line per quantity. `ground_truth/claims_gate.R` runs the second file as a program, not as text, and halts the run unless three things hold: the number of printed claims equals the 407 extraction rows

requiring one, the two identifier sets are equal in both directions, and every printed value agrees with the ground truth at the precision the page uses. A claim block that errored, printed nothing, or silently read the wrong file would fail that gate.

The independence is the point. The gate found two defects in this repository that no number-against-number check could: `appendix_c_margins_turnout.R` was labelling Experiment 1's rows with the dependent variable's name rather than the outcome it was passed, so a filter on turnout silently dropped one of four experiments, and the appendix D pooled interaction effects, the section 5.5 pooled turnout effect and footnote 3's cost per vote had no counterpart in `output/` at all.

5 Coverage

The article and its appendix print 481 numbers. 400 of them are produced by the pipeline and compared cell by cell; 7 are judged as claims rather than as values; the other 74 are facts about how the experiments were run that no deposited file records.

Table 4: Every published float, with the number of cells it prints and how many reproduce. Coverage is complete: no float has a cell without a row.

Float	Published cells	Archive comparable	Archive matching	Rewrite comparable	Rewrite matching
Table 2	16	16	16	16	16
Table 3	18	18	18	18	18
Table 4	18	18	4	18	18
Table 5	18	18	18	18	18
Table 6	18	18	18	18	18
Table 7	20	20	20	20	20
Figure 2	45	45	45	45	45
Table C.1	34	34	34	34	34
Table C.2	34	34	6	34	34
Table C.3	34	34	34	34	34
Table C.4	34	34	34	34	34
Table D.5	56	56	56	56	56

6 Number-by-number comparison

Of the 481 published values, 378 can be compared against what a deposited script prints. 335 of those match and 43 do not. Every failure is listed in full below, and every one is the Experiment 2 seed.

Table 5: Every published value a deposited script fails to reproduce on current R. The rewrite reproduces all of them, which is the signature of environment drift rather than of an error in the deposit.

Location	Claim	Paper	Archive	Rewrite	Locus
Section 5.2, p. 147	text_r2_ri_p	0.90	0.79	0.90	environment
Table 4	t4_m1_adjacent_coef	0.012	0.137	0.012	environment
Table 4	t4_m1_adjacent_se	0.046	0.072	0.046	environment
Table 4	t4_m1_constant_coef	0.659	0.543	0.659	environment
Table 4	t4_m1_constant_se	0.039	0.068	0.039	environment
Table 4	t4_m1_r2	0.001	0.130	0.001	environment
Table 4	t4_m1_treated_coef	0.009	0.130	0.009	environment
Table 4	t4_m1_treated_se	0.054	0.080	0.054	environment
Table 4	t4_m2_adjacent_coef	0.004	0.028	0.004	environment
Table 4	t4_m2_adjacent_se	0.045	0.035	0.045	environment
Table 4	t4_m2_constant_coef	0.287	0.146	0.287	environment
Table 4	t4_m2_constant_se	0.131	0.101	0.131	environment
Table 4	t4_m2_r2	0.253	0.588	0.253	environment
Table 4	t4_m2_treated_coef	-0.014	0.019	-0.014	environment
Table 4	t4_m2_treated_se	0.057	0.053	0.057	environment
Table C.2	c2_margin_m1_adjacent_coef	21.436	34.970	21.436	environment
Table C.2	c2_margin_m1_adjacent_se	12.565	14.896	12.565	environment
Table C.2	c2_margin_m1_constant_coef	32.415	22.642	32.415	environment
Table C.2	c2_margin_m1_constant_se	9.733	12.294	9.733	environment
Table C.2	c2_margin_m1_r2	0.048	0.087	0.048	environment
Table C.2	c2_margin_m1_treated_coef	24.673	21.495	24.673	environment
Table C.2	c2_margin_m1_treated_se	17.196	19.694	17.196	environment
Table C.2	c2_margin_m2_adjacent_coef	5.563	14.440	5.563	environment
Table C.2	c2_margin_m2_adjacent_se	12.363	9.834	12.363	environment
Table C.2	c2_margin_m2_constant_coef	-30.993	-20.217	-30.993	environment
Table C.2	c2_margin_m2_constant_se	15.629	16.556	15.629	environment
Table C.2	c2_margin_m2_r2	0.501	0.556	0.501	environment
Table C.2	c2_margin_m2_treated_coef	4.427	-2.839	4.427	environment
Table C.2	c2_margin_m2_treated_se	14.470	14.972	14.470	environment
Table C.2	c2_turnout_m1_adjacent_coef	33.191	44.038	33.191	environment
Table C.2	c2_turnout_m1_adjacent_se	15.131	19.682	15.131	environment
Table C.2	c2_turnout_m1_constant_coef	92.776	83.329	92.776	environment
Table C.2	c2_turnout_m1_constant_se	11.927	16.506	11.927	environment
Table C.2	c2_turnout_m1_r2	0.067	0.095	0.067	environment
Table C.2	c2_turnout_m1_treated_coef	38.407	37.614	38.407	environment
Table C.2	c2_turnout_m1_treated_se	23.037	26.008	23.037	environment
Table C.2	c2_turnout_m2_adjacent_coef	8.012	14.943	8.012	environment
Table C.2	c2_turnout_m2_adjacent_se	11.276	14.808	11.276	environment
Table C.2	c2_turnout_m2_constant_coef	-43.402	-42.177	-43.402	environment
Table C.2	c2_turnout_m2_constant_se	15.695	16.529	15.695	environment
Table C.2	c2_turnout_m2_r2	0.695	0.702	0.695	environment
Table C.2	c2_turnout_m2_treated_coef	9.029	5.985	9.029	environment
Table C.2	c2_turnout_m2_treated_se	13.372	15.442	13.372	environment

The complete row-by-row table, all 481 of them, is `ground_truth/green_etal_2016_ground_truth.csv`.

7 Maintained rewrite

The rewrite lives in `maintained/`: seventeen scripts covering four cleaning steps, the six published tables, the appendix margin and turnout tables, appendix Table D.5, Figure 2,

footnote 3's cost calculation, the randomization inference, and the in-text claims. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used. An eighteenth script, `text_exp1_merged_shares.R`, is the exception by design: it recomputes everything downstream of Experiment 1's vote shares for errata entry 1, and leaves the published objects alone.

7.1 Architecture

Each experiment's analysis reads the archive's canonical `exp_N.RData` object, which carries both the analysis frame and that experiment's 10,000-column permutation matrix. The `clean_expN.R` scripts rebuild the same frames from the raw deposited files in tidy form and are the readable account of how each variable is constructed, but no published number is taken from them. That separation is deliberate and load-bearing: `clean_exp2.R` redraws the observed assignment with the same seed the archive uses, and therefore selects the wrong column on any current R.

Weighted models use `estimatr::lm_robust(se_type = "HC2")` in place of `lm()` followed by a `coeftest(fit, vcovHC(fit, type = "HC2"))` wrapper, which is the same estimator with the same standard errors. Pooling uses `metafor::rma(method = "FE")` in place of `rmeta::meta.summaries()`. The randomization inference reimplements the archive's `get_f()` as `f_treat()` in `helpers.R` and loops over the deposited permutation matrices, computing all four experiments rather than carrying any p-value forward as a constant.

7.2 Deprecated patterns replaced

Table 6: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘setwd(”“)‘	‘here::here()‘
‘lm()‘ + ‘coeftest(vcovHC, ’HC2’)‘	‘estimatr::lm_robust(se_type = ”HC2”)‘
‘rmeta::meta.summaries(method = ’fixed’)‘	‘metafor::rma(method = ”FE”)‘
‘stargazer‘ + ‘sink()‘	‘write_csv()‘ to ‘output/‘
‘xtable‘ + ‘print.xtable‘	‘write_csv()‘ to ‘output/‘
‘within()‘ blocks over base indexing	‘mutate()‘ paragraphs
‘merge()‘	‘left_join()‘
‘multiplot()‘ (grid-based)	‘gridExtra::arrangeGrob()‘
‘pdf()‘ / ‘png()‘ devices	‘ggsave()‘
‘library(beepr)‘ + ‘beep()‘	(omitted)
magrittr pipe	native pipe

8 Randomization inference

The article reports one randomization inference p-value per experiment, for vote share, in sections 5.1 through 5.4. All four reproduce.

Table 7: Randomization inference p-values for the joint null of no direct and no indirect effect, 10,000 permutations each. The article states the vote share p-values only.

Experiment	Outcome	Rewrite	Published
Experiment 1	share	0.2225	0.22
Experiment 1	margin	0.1573	
Experiment 1	turnout	0.5261	
Experiment 2	share	0.8974	0.90
Experiment 2	margin	0.8928	
Experiment 2	turnout	0.6967	
Experiment 3	share	0.0208	0.02
Experiment 3	margin	0.2633	
Experiment 3	turnout	0.3322	
Experiment 4	share	0.7714	0.77
Experiment 4	margin	0.8770	
Experiment 4	turnout	0.8509	

The margin and turnout p-values are computed here for completeness and appear nowhere in the article or its appendix, so they have no ground-truth row: the ground truth records what the paper claims, and a quantity the paper never states is not a claim. Producing them is cheap: all twelve p-values together, 40,000 permutations and 120,000 pairs of weighted models, take under two minutes.

Every p-value here is computed rather than carried. The permutation matrices are already inside the `exp_3.RData` and `exp_4.RData` objects the table scripts load, so no additional file is read, and the two loops together run in under a minute. A number typed into a script is not a number the script produces.

9 Figure 2 verification

Figure 2 traces how an agnostic, an optimistic and a skeptical observer update a prior about the direct effect as the four experiments arrive, and each of its fifteen panels is labelled with the posterior mean, the posterior standard deviation and the posterior probability that the effect is positive. Those forty-five numbers are the figure's content, and all forty-five reproduce.

Table 8: Figure 2 panel labels, transcribed from page 149 of the article, against the maintained rewrite.

Observer	Panel	Quantity	Paper	Rewrite	Match
Agnostic	Prior	mean	0.000	0.000	1
	Prior	pr(ATE > 0)	0.500	0.500	1
	Prior	sd	0.050	0.050	1
	Update 1	mean	0.022	0.022	1
	Update 1	pr(ATE > 0)	0.915	0.915	1
	Update 1	sd	0.016	0.016	1
	Update 2	mean	0.019	0.019	1
	Update 2	pr(ATE > 0)	0.895	0.895	1
	Update 2	sd	0.016	0.016	1
	Update 3	mean	0.019	0.019	1
	Update 3	pr(ATE > 0)	0.993	0.993	1
	Update 3	sd	0.008	0.008	1
	Update 4	mean	0.016	0.016	1
	Update 4	pr(ATE > 0)	0.988	0.988	1
	Update 4	sd	0.007	0.007	1
Optimist	Prior	mean	0.050	0.050	1
	Prior	pr(ATE > 0)	0.841	0.841	1
	Prior	sd	0.050	0.050	1
	Update 1	mean	0.027	0.027	1
	Update 1	pr(ATE > 0)	0.955	0.955	1
	Update 1	sd	0.016	0.016	1
	Update 2	mean	0.024	0.024	1
	Update 2	pr(ATE > 0)	0.941	0.941	1
	Update 2	sd	0.016	0.016	1
	Update 3	mean	0.020	0.020	1
	Update 3	pr(ATE > 0)	0.996	0.996	1
	Update 3	sd	0.008	0.008	1
	Update 4	mean	0.017	0.017	1
	Update 4	pr(ATE > 0)	0.991	0.991	1
	Update 4	sd	0.007	0.007	1
Skeptic	Prior	mean	0.000	0.000	1
	Prior	pr(ATE > 0)	0.500	0.500	1
	Prior	sd	0.010	0.010	1
	Update 1	mean	0.006	0.006	1
	Update 1	pr(ATE > 0)	0.768	0.768	1
	Update 1	sd	0.009	0.009	1
	Update 2	mean	0.006	0.006	1
	Update 2	pr(ATE > 0)	0.754	0.754	1
	Update 2	sd	0.009	0.009	1
	Update 3	mean	0.012	0.012	1
	Update 3	pr(ATE > 0)	0.977	0.977	1
	Update 3	sd	0.006	0.006	1
	Update 4	mean	0.011	0.011	1
	Update 4	pr(ATE > 0)	0.966	0.966	1
	Update 4	sd	0.006	0.006	1

The rewrite updates on the fitted coefficients at full precision, reading them from `maintained/output/table_7_pooled.csv`, which is what the original does. Retyping the four direct effects at the three decimal places Table 7 prints would carry a rounding into a calculation rather than into a display.

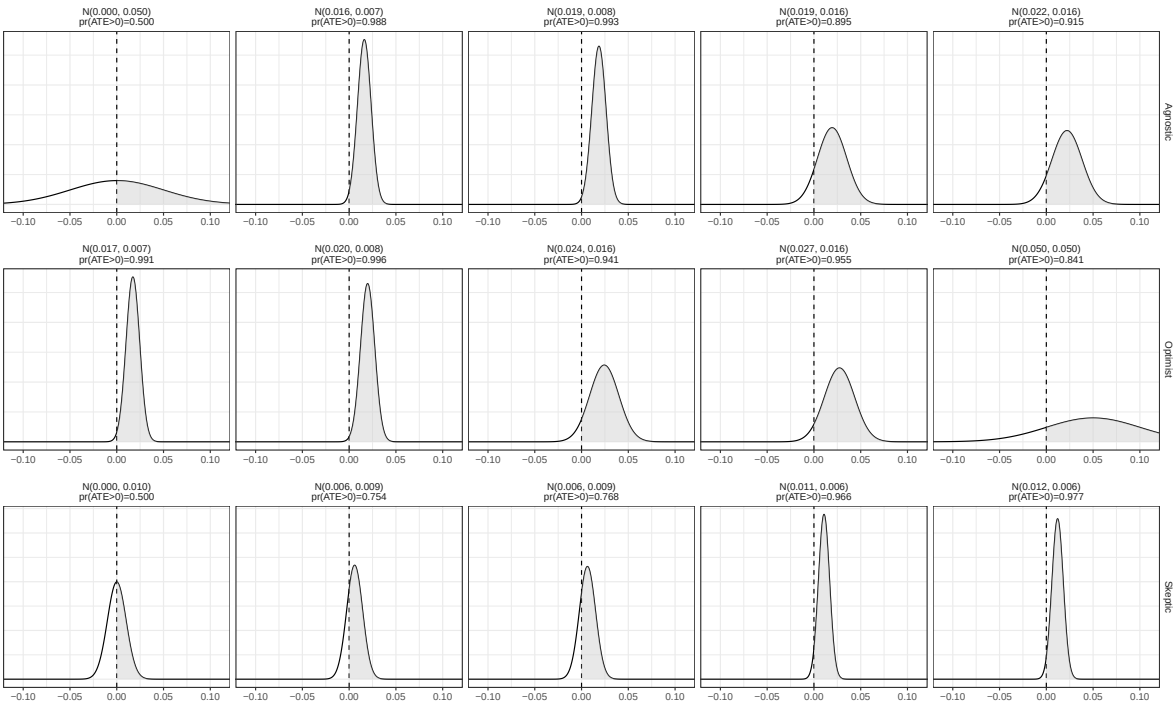


Figure 1: Figure 2 as reproduced by the maintained rewrite.

10 Maintained rewrite verification

Every published value with a counterpart in the pipeline carries `match_rewrite = 1`: **400** of 400, with 0 differing. The prose is where the coverage was thinnest and is now complete, so it is worth setting out separately.

Table 9: Every quantity the article states in prose, in a footnote, or in the appendix text, against the maintained rewrite.

Location	Claim	Paper	Rewrite	Verdict
Appendix B	appb_e1_untreatable	5	5	matches
Appendix B	appb_e2_permutations	10000	10000	matches
Appendix B	appb_e3_permutations	10000	10000	matches
Appendix B	appb_e3_precincts	131	131	matches
Appendix B	appb_e3_treated	30	30	matches
Appendix B	appb_e4_permutations	10000	10000	matches
Appendix B	appb_e4_precincts	88	88	matches
Appendix B	appb_e4_treated	20	20	matches
Appendix D	appd_e4_negative_interactions	1	1	holds
Appendix D	appd_no_precise_interactions	1	1	holds
Appendix D	appd_pooled_direct_interaction	0.009	0.009	matches
Appendix D	appd_pooled_direct_interaction_se	0.007	0.007	matches

Table 9: Every quantity the article states in prose, in a footnote, or in the appendix text, against the maintained rewrite. *(continued)*

Location	Claim	Paper	Rewrite	Verdict
Appendix D	appd_pooled_indirect_interaction	0.007	0.007	matches
Appendix D	appd_pooled_indirect_interaction_se	0.007	0.007	matches
Appendix D	appd_three_of_four_positive	3	3	holds
Footnote 1, p. 144	text_fn1_e1_treated_assigned	23	23	matches
Footnote 1, p. 144	text_fn1_e4_treated_assigned	20	20	matches
Footnote 3, p. 149	text_fn3_ci_high	13.71	13.17	holds
Footnote 3, p. 149	text_fn3_ci_level	95	95	matches
Footnote 3, p. 149	text_fn3_ci_low	1.70	1.70	holds
Footnote 3, p. 149	text_fn3_cost_direct	3.18	3.18	matches
Footnote 3, p. 149	text_fn3_cost_total	1.69	1.69	matches
Footnote 3, p. 149	text_fn3_direct_points	1.7	1.7	matches
Footnote 3, p. 149	text_fn3_total_cost	13045	13045	matches
Footnote 3, p. 149	text_fn3_total_turnout	241613	241613	matches
Section 3, p. 144	text_design_e1_treatable	88	88	matches
Section 3, p. 144	text_design_e2_initial	128	128	matches
Section 3, p. 144	text_design_e2_restricted	69	69	matches
Section 3, p. 144	text_design_e3_precincts	131	131	matches
Section 3, p. 144	text_design_e4_precincts	88	88	matches
Section 4, p. 147	text_ri_permutations	10000	10000	matches
Section 5.1, p. 147	text_r1_direct_m1	2.5	2.5	matches
Section 5.1, p. 147	text_r1_direct_m1_se	2.7	2.7	matches
Section 5.1, p. 147	text_r1_direct_m2	2.5	2.5	matches
Section 5.1, p. 147	text_r1_direct_m2_se	1.7	1.7	matches
Section 5.1, p. 147	text_r1_ri_p	0.22	0.22	matches
Section 5.1, p. 147	text_r1_spillover_m1	3.7	3.7	matches
Section 5.1, p. 147	text_r1_spillover_m1_se	2.7	2.7	matches
Section 5.1, p. 147	text_r1_spillover_m2	1.8	1.8	matches
Section 5.1, p. 147	text_r1_spillover_m2_se	1.6	1.6	matches
Section 5.2, p. 147	text_r2_direct_m1	0.9	0.9	matches
Section 5.2, p. 147	text_r2_direct_m2	1.4	1.4	matches
Section 5.2, p. 147	text_r2_ri_p	0.90	0.90	matches
Section 5.3, p. 148	text_r3_direct_m2	1.8	1.8	matches
Section 5.3, p. 148	text_r3_ri_p	0.02	0.02	matches
Section 5.3, p. 148	text_r3_spillover_m2	1.8	1.8	matches
Section 5.4, p. 148	text_r4_direct_m2	1.2	1.2	matches
Section 5.4, p. 148	text_r4_ri_p	0.77	0.77	matches
Section 5.4, p. 148	text_r4_spillover_m2	2.0	2.0	matches
Section 5.5, p. 148	text_turnout_no_effect	1	1	holds
Section 5.5, p. 148	text_turnout_pooled_direct	7.2	7.2	matches
Section 5.5, p. 148	text_turnout_pooled_direct_se	9.5	9.5	matches
Section 6, p. 148	text_agnostic_pr	98.8	98.8	matches
Section 6, p. 148	text_optimist_pr	0.991	0.991	matches
Section 6, p. 148	text_pooled_direct	1.7	1.7	matches
Section 6, p. 148	text_pooled_direct_se	0.7	0.7	matches
Section 6, p. 148	text_pooled_indirect	1.5	1.5	matches
Section 6, p. 148	text_pooled_indirect_se	0.6	0.6	matches
Section 6, p. 148	text_pooled_just_over_one	1	1	holds
Section 6, p. 148	text_prior_diffuse_sd	5	5	matches
Section 6, p. 148	text_prior_skeptic_sd	1	1	matches
Section 6, p. 148	text_skeptic_pr	0.966	0.966	matches

Two of these need a word. Footnote 3's cost per vote is computed by the article from the pooled direct effect as Table 7 prints it, at three decimals, so $13,045/(241{,}613 \times 0.017) = \$3.18$ is the article's own arithmetic and reproduces exactly; at the pooled estimate's full precision the same calculation gives \$3.26. And the footnote's 95 per cent interval is drawn in the deposit from 10,000 normal variates with no seed, so no single number is the published one. Over 200 seeds the lower endpoint runs from \$1.66 to \$1.73 and the upper from \$11.79 to \$15.19, and the published \$1.70 and \$13.71 both sit inside those ranges.

11 Errata

`errata.qmd` renders to `green_etal_2016_errata.pdf` at the root of this repository and writes its entries to `errata_entries.csv`. The note lists 23 corrections and one unresolved citation, and none changes the paper's conclusions.

Entry 1 is the only one that moves an estimate. Line 68 of `Experiment_1_Analysis.R` sums the share columns across the precincts merged for assignment after the shares are computed, so the four merged units, all in the adjacent arm, carry the sum of two shares. With each merged unit's shares taken from its summed vote counts, Table 3's spillover estimate goes from 0.037 to 0.007 in Model 1 and from 0.018 to 0.019 in Model 2, so covariate adjustment raises the estimate where the article says it falls. Table 7's pooled estimates hold at the precision it prints. `maintained/text_exp1_merged_shares.R` writes the corrected values to `maintained/output/text_exp1_merged_shares.csv` and leaves the published objects alone, so the rewrite still reproduces the page. The entry's 38 ground-truth rows therefore keep their matching verdicts and carry `defect_locus = archive`: the page reports the deposit faithfully, and the deposit's code is wrong.

Entries 2 and 24 are mirror errors of description: the notes to Tables 3 to 6 call past vote shares margins, and the notes to Tables C.1 to C.4 call past vote margins shares. No estimate changes. Table 5's missing `R2` label and the appendix's jump from Table C.4 to Table D.5, which this report formerly recorded as falling short of errata, are entries 5 and 22. The remaining entries correct wording, spelling, and references.

The extraction covers all 481 numbers the article and its appendix print, and no row carries `defect_locus = paper_internal`: none of those numbers contradicts the article's own tables. The 43 values a deposited script no longer reproduces are a property of R rather than of the paper, and the paper's numbers are the correct ones.

12 R environment

Item	Value
R version	4.6.1

Platform	aarch64-apple-darwin23
Date run	2026-11-05

Table 11: Package versions used for the run behind this report.

Package	Version
estimatr	2.0.1.9000
metafor	5.0.1
dplyr	1.2.1
ggplot2	4.0.3
tidyr	1.3.2
purrr	1.2.2
here	1.0.2
gridExtra	2.3